DELTASAT TOOLKIT

USER MANUAL

Intertidal extent & elevation of the Meghna estuary

v1.0 - October 2026

1 What the toolkit does

DeltaSAT maps the intertidal area of the Meghna estuary from Landsat satellite imagery and hourly gauge records. You draw a region on a satellite map and press one button; the toolkit computes the yearly intertidal extent, the lifespan of each intertidal patch, and the relative elevation of the intertidal surface.

- One analysis: intertidal extent & elevation.
- Nothing to upload - imagery (Landsat 5/8 via the Google Earth Engine Python API) and water levels (10 gauges of the GBM hydro-morphological database) are fetched automatically.
- Everything runs on your own machine; results appear in the browser.

2 What you need

- ☐ Python 3.10 or newer
- ☐ An internet connection
- ☐ A Google Earth Engine account (free for research - code.earthengine.google.com). Only needed for real results.
- ☐ The gbm-toolkit folder, placed next to the demo folder

3 One-time setup

1 Install the Python packages

```
cd demo
pip install -r requirements.txt
```

2 Log in to Earth Engine (once per machine)

```
earthengine authenticate
```

3 Check the folder layout

Both folders must sit side by side:

```
DeltaSAT-/
  demo/          <- the web app (this manual's subject)
  gbm-toolkit/   <- the gbm_delta analysis package
```

4 Start the toolkit

1 Run the launcher

```
cd demo
python run.py
```

This starts the server AND opens the toolkit in your browser (http://localhost:8000/toolkit.html). Keep the terminal open while you work - closing it stops the toolkit.

2 Read the status line (top of the results panel)

Status line	Meaning
connected - live Earth Engine	Ready. Analyses will use real satellite data.
connected - MOCK mode	Works, but figures are placeholders. Earth Engine is not authenticated - run 'earthengine authenticate' and restart.
unreachable	The page was opened as a local file. Start the backend with 'python run.py' and open http://localhost:8000/toolkit.html.

5 Run an analysis

1. **Draw the region**
   Click ’Draw ROI’ and drag a rectangle on the satellite map. Keep it modest (10-30 km across); large regions take much longer.
2. **Set the period**
   Start/end year (Landsat 5: 1984-2012, Landsat 8: 2013 onward) and the maximum cloud cover (20% is a good default).
3. **Press ’Run analysis’**
4. **Watch the progress**
   The status line shows ’Submitting...’, then the backend log lists each step (Landsat export, intertidal detection, tidal datums, elevation).
5. **Read the results**
   The output figure appears in the panel; the backend log underneath explains everything that was computed - or skipped, and why.

Gauge stations of the hydro database appear on the map automatically. Stations inside your box are highlighted - their records feed the elevation step.

6 Reading the results

Output	What it shows
Intertidal extent	Where intertidal flats existed, year by year.
Lifespan	How many years each intertidal patch survived (transition-matrix).
Elevation	Relative height (m) of the intertidal surface, derived from water lines and gauge tidal datums (MHSW/MLSW).
Backend log	Every processing step with its status. Skipped steps state the reason here.

7 Modes

By default the toolkit picks its mode automatically: it uses live Earth Engine when you are authenticated, and falls back to MOCK placeholders otherwise. You can force either:

```
MOCK_MODE=1 python run.py                # placeholders, no GEE needed
MOCK_MODE=0 EE_PROJECT="my-project" python run.py   # force live GEE
```